

June 11th 2024

Graphene synthesis project update

1. Biochar synthesis

- Overall seven syntheses finished with hemp fibers from BAFA neu GmbH
 - three syntheses done with 0.02M sulfuric acid
 - first product not representative, not used further
 - second and third batch converted (see below)
 - four syntheses done with 0.10M sulfuric acid
 - very similar observations and outputs
 - products of these have been combined/homogenized
 - part of the combined lots has been converted
- Canadian Rockies Hemp arrived and cleared customs
 - ten synthesis batches finished (16.3 g product in total), seven of these (11.2 g) were combined and homogenized
- (Re)-Activation of larger vessel in progress

2. Biochar analysis

- Sample of combined biochar (milled, MB2946X_gem) analyzed by SEM and Raman for reference

3. Graphene synthesis

- two experiments finished with steep heating ramp (20-30 °C/min)
 - biochar ground manually for first experiment
 - laboratory mill used for second experiment
 - products not yet characterized
 - no serious handling issues
- two experiments performed with much gentler heating ramp (3 °C/min)
 - better handling but no other apparent difference
- 2 g of combined biochar milled with 2 g of KOH and used for several batches on 0.25 g scale
 - 800 °C, 1 h: minimal yield
 - 800 °C, 2 h: minimal yield
 - 800 °C, 3 h: no output
 - ➔ Inertion apparently insufficient and material likely incinerated
 - 800 °C, 3 h repeat, much higher Ar stream: 38 mg output obtained

4. Graphene analysis

- SEM and Raman results available for
 - TB1133, MB2946A (step 2 products)
 - 900561-500MG (Sigma-Aldrich reference)
- Discussion with potential service provider (TEM, BET) in progress

5. Currently open questions

Experiment overview

Experiment	Reactor	Raw material	Acid	T	t	pressure	Wash	Output	Experiment	Qty	Base	grinding	Gas	T (rate)	T (max)	t	Wash	Drying	Output	
MB2928	AV1	6.0 g BAFA neu Hemp Fibre VF	100.0 g 0.19% 0.02 M Sulfuric acid	180 °C	22.5 h	8.7 bar -> 10.4 bar	742 g Water	2.6 g	->											
Comment	Fibres not completely covered by liquid																			
MB2933	AV1	6.1 g BAFA neu Hemp Fibre VF	174.0 g 0.19% 0.02 M Sulfuric acid	180 °C	24 h	8.7 bar -> 9.5 bar	647 g Water	1.2 g	->	TB1131	1.2 g	1.2 g KOH (90%)	manual	Ar	20-27 °C/min	790 °C	1 h	20 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.16 g
Comment	dark crusts on glassware, metal and liquid surface removed before filtration																			
MB2935	AV1	6.0 g BAFA neu Hemp Fibre VF	172.9 g 0.19% 0.02 M Sulfuric acid	180 °C	23 h	8.8 bar -> 9.5 bar	500 g Water	1.3 g	->	TB1132	1.3 g	1.3 g KOH (90%)	mill (2 min)	Ar	20-29 °C/min	792 °C	1 h	20 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.3 g
Comment	dark crusts on glassware, metal and liquid surface removed before filtration																			
MB2936	AV1	6.0 g BAFA neu Hemp Fibre VF	172.4 g 0.96% 0.10 M Sulfuric acid	180 °C	23 h	8.8 bar -> 10.6 bar	589 g Water	0.8 g	->	TB1133	1.0 g	1.0 g KOH (90%)	mill (1 min)	Ar	3 °C/min	785 °C	1 h	20 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.3 g
Comment	dark crusts on glassware, metal and liquid surface removed before filtration																			
MRa225	AV1	6.0 g BAFA neu Hemp Fibre VF	153.8 g 0.96% 0.10 M Sulfuric acid	180 °C	24 h	? -> 10.5 bar	500 g Water	0.9 g	->	TB1134	1.0 g	1.0 g KOH (90%)	mill (2 min)	Ar	3 °C/min	771 °C	1 h	20 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.3 g
Comment	dark crusts on glassware, metal and liquid surface removed before filtration																			
MRa228	AV1	6.3 g BAFA neu Hemp Fibre VF	154.0 g 0.96% 0.10 M Sulfuric acid	180 °C	22.5 h	8.7 bar -> 11.0 bar	500 g Water	0.9 g	->											
Comment	dark crusts on glassware, metal and liquid surface removed before filtration																			
MRa231	AV1	6.1 g BAFA neu Hemp Fibre VF	168.8 g 0.96% 0.10 M Sulfuric acid	180 °C	23.5 h	8.0 bar -> 10.5 bar	500 g Water	0.8 g	->											
Comment	dark crusts on glassware, metal and liquid surface removed before filtration																			
MB2937	AV1	9.4 g Canadian Rockies Hemp	179.3 g 0.96% 0.10 M Sulfuric acid	180 °C	24 h	8.6 bar -> 10.9 bar	569 g Water	1.6 g	->	MB2946	1.0 g	1.0 g KOH (90%)	mill (1 min)	Ar	5 °C/min	770 °C	1 h	22 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	0.3 g
Comment	very low amount of foreign material observed and removed during filtration																			
MB2938	AV1	8.9 g Canadian Rockies Hemp	175.0 g 0.96% 0.10 M Sulfuric acid	180 °C	23.5 h	9.2 bar -> 13.2 bar	577 g Water	1.4 g	->	MB2947A	0.25 g	0.25 g KOH (90%)	mill (1 min)	Ar	3 °C/min	803 °C	1 h	10 g 10% HCl (+ water)	40 °C N2 stream 210 mbar	< 0.01 g
Comment	very low amount of foreign material observed and removed during filtration																			
MB2940	AV1	9.2 g Canadian Rockies Hemp	177.1 g 0.96% 0.10 M Sulfuric acid	180 °C	22.5 h	8.7 bar -> 10.6 bar	555 g Water	1.5 g	->	MB2947B	0.25 g	0.25 g KOH (90%)	mill (1 min)	Ar	3 °C/min	804 °C	2 h	10 g 10% HCl (+ water)	40 °C N2 stream 210 mbar	< 0.01 g
Comment	very low amount of foreign material observed and removed during filtration																			
MB2942	AV1	8.9 g Canadian Rockies Hemp	177.6 g 0.96% 0.10 M Sulfuric acid	180 °C	23 h	8.3 bar -> 13.5 bar	578 g Water	1.6 g	->	MB2947C	0.25 g	0.25 g KOH (90%)	mill (1 min)	Ar	3 °C/min	806 °C	3 h	Batch discarded		
Comment	low amount of foreign material observed and removed during filtration																			
MB2943	AV1	9.8 g Canadian Rockies Hemp	178.2 g 0.96% 0.10 M Sulfuric acid	180 °C	24 h	8.3 bar -> 12.7 bar	611 g Water	1.7 g	->	MB2947D	0.25 g	0.25 g KOH (90%)	mill (1 min)	Ar	3 °C/min	810 °C	3 h	10 g 10% HCl (+ water)	100 °C N2 stream (atm. Pressure)	38 mg
Comment	low amount of foreign material observed and removed during filtration																			
MB2944	AV1	9.0 g Canadian Rockies Hemp	178.4 g 0.96% 0.10 M Sulfuric acid	180 °C	24 h	8.4 bar -> 9.7 bar	593 g Water	1.8 g	->											
Comment	low amount of foreign material observed and removed during filtration																			
MB2945	AV1	9.1 g Canadian Rockies Hemp	177.7 g 0.96% 0.10 M Sulfuric acid	180 °C	24 h	8.7 bar -> 9.8 bar	6 g Water	1.6 g	->											
Comment	low amount of foreign material observed and removed during filtration																			

Experiment	Reactor	Raw material	Acid	T	t	pressure	Wash	Output	Experiment	Qty	Base	grinding	Gas	T (rate)	T (max)	t	Wash	Drying	Output
MB2948	AV1	9.4 g Canadian Rockies Hemp	175.1 g 0.96% 0.10 M Sulfuric acid	180 °C	25 h	9.0 bar -> 12.6 bar	475 g Water	1.7 g											
Comment	low amount of foreign material observed and removed during filtration																		
MB2949	AV1	9.2 g Canadian Rockies Hemp	174.3 g 0.96% 0.10 M Sulfuric acid	180 °C	23.5 h	8.4 bar -> 10.2 bar	538 g Water	1.8 g											
Comment	low amount of foreign material observed and removed during filtration																		
MB2950	AV1	9.1 g Canadian Rockies Hemp	170.5 g 0.96% 0.10 M Sulfuric acid	180 °C	23 h	-> 11.1 bar	534 g Water	1.6 g											
Comment	low amount of foreign material observed and removed during filtration																		